# Real-Time DeepJSCC-Q on FPGA

Wireless image transmission on two Zynq boards

## 30 fps

256 × 256 RGB
18,020 frames in 600 s

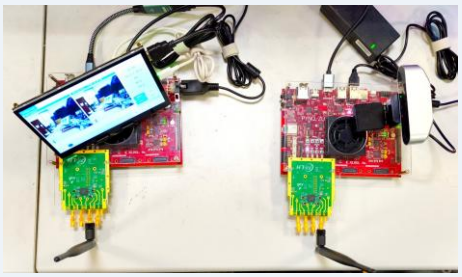
## 67.5 ms

PL frame latency
Frames overlap in the pipeline

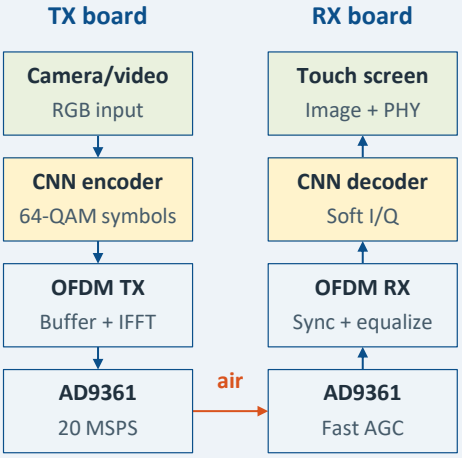
## +2.8 dB

High-SNR PSNR gain over SSCC
DIV2K 0802; equal symbol budget

## Two-board wireless link



RX (left), TX (right).
2 × PYNQ-ZU + 2 × AD-FMCOMMS3.



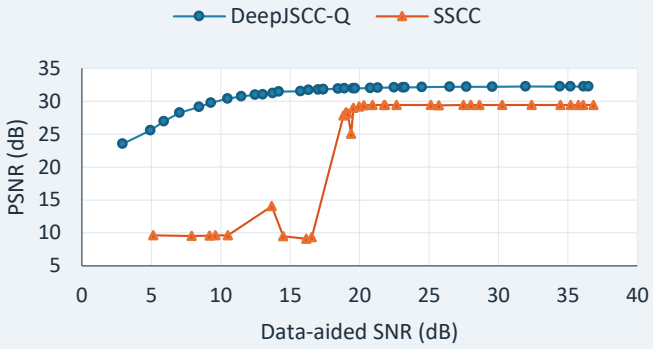

**32,768 complex symbols/image**
**683 OFDM symbols/frame**
**20 MSPS PHY sample rate**

Soft I/Q enters the decoder without hard decisions. Ethernet carries control and the reference preview.

PL latency excludes app/display.
Transfer includes buffers + RX PHY.

## Measured quality over the air

DIV2K 0802; same PHY and 32,768 symbols/image.
SSCC: JPEG + 802.11n LDPC + 64-QAM.

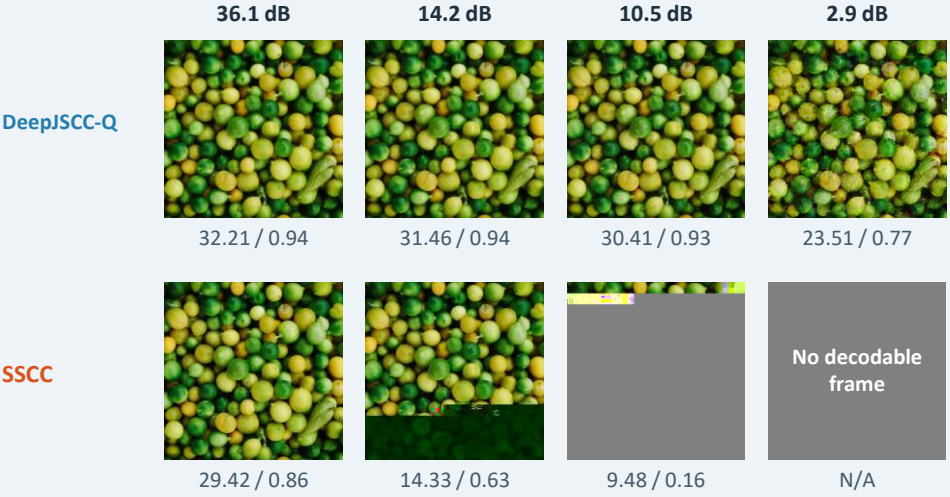



**Original**



256 × 256
RGB

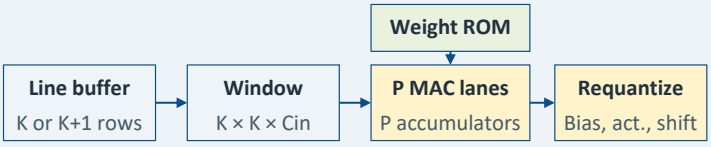
### Reconstruction at four channel conditions




Matched TX attenuation; column SNR of the DeepJSCC-Q frames.
Curve: window means. Images: frame PSNR (dB) / SSIM.
Decode failures are scored as mid-gray.

## Streaming FPGA implementation

250 MHz, W8A12, 1.15 GMAC/frame



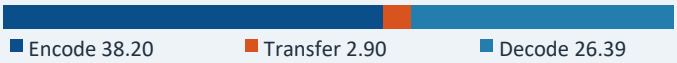

Layer engines form a pipeline; P meets 30 fps.
On-chip buffers avoid full intermediate feature maps.

### Routed TX / RX totals on xczu5eg

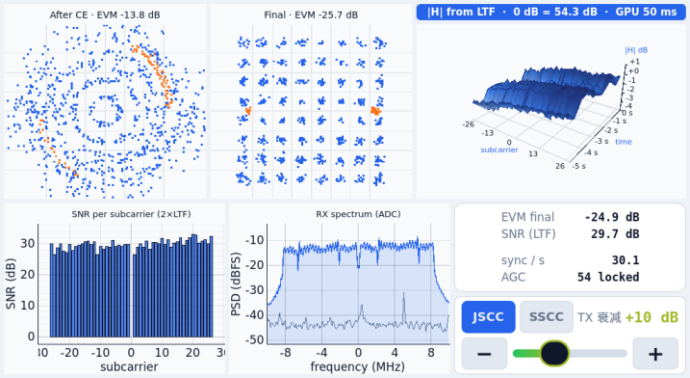
| Design | LUT | BRAM36 | URAM | DSP |
|---|---|---|---|---|
| **TX** | 54,241 | 104.5 | 10 | 243 |
| **RX** | 63,699 | 65.5 | 7 | 133 |

Measured rails: TX 3.07 W / RX 3.44 W
PL + PS + DDR; excludes RF and input conversion losses.

## PL frame latency: 67.5 ms




## Live PHY diagnostics




DeepJSCC: TCCN 2019. DeepJSCC-Q: JSAIT 2022. Developed with Claude Code and Codex under human direction.